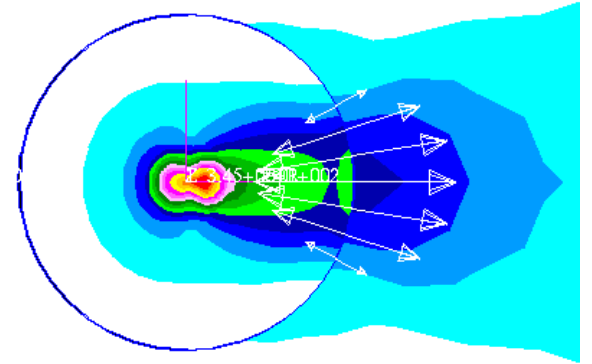




“Linear Gap” New Capability Seminar

October 2001

Linear Gaps

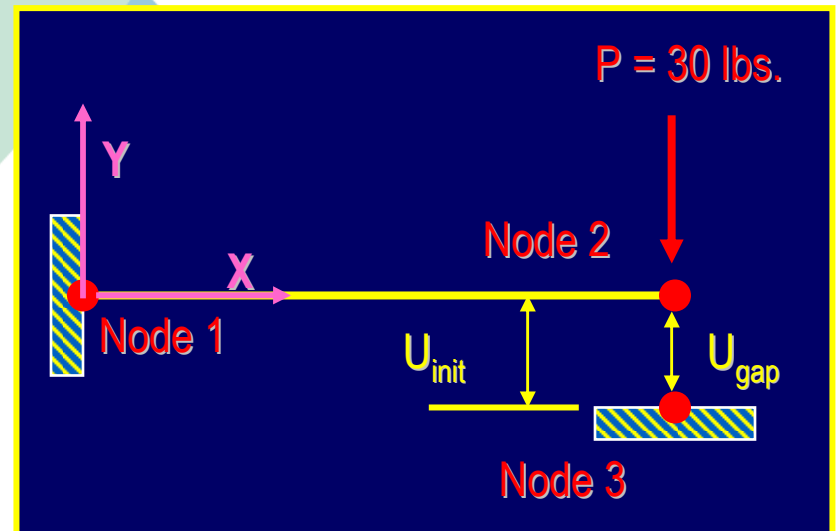


- What is a “linear gap”?
- Why do I want linear gaps?
- What is theory/implementation?
- Patran interface
- Examples
- Limitations
- Conclusions

What is a “linear gap”?

- The linear gap is a contact element in its simplest form:
 - DOF-to-DOF gaps with initial openings
 - Linear, therefore does not include:
 - friction
 - large displacements
 - material non-linearity

Uses iterative solver to determine open/closed status.

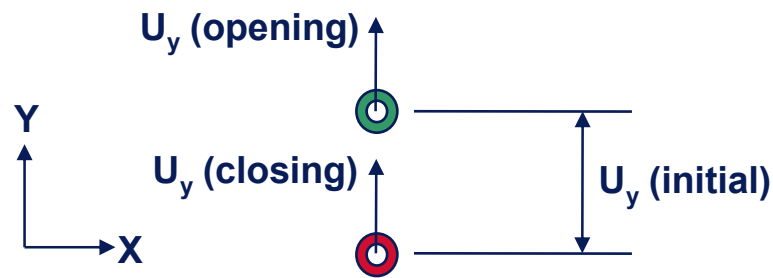
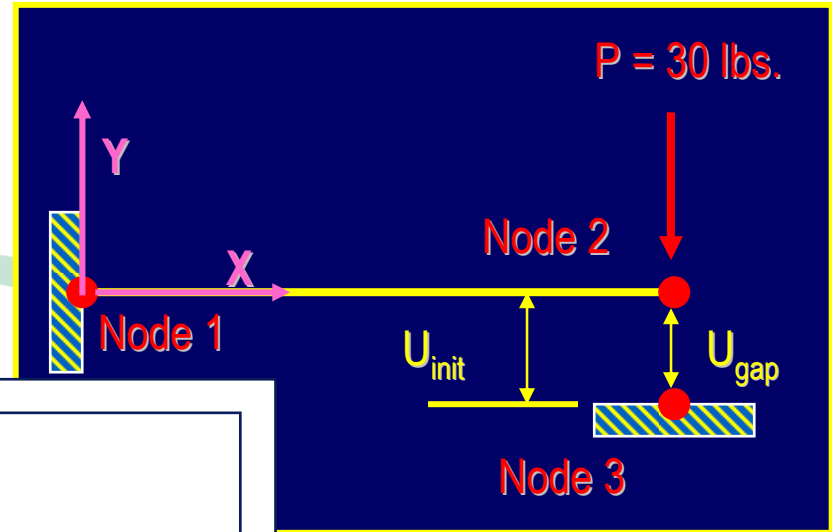


Why do I want linear gaps?

- **Successful analysis will use proper boundary conditions and load paths.**
- **Linear gaps can solve a large class of problems**
 - Heel-toe interaction
 - Hole bearing
 - Simple contacts
 - Bridge Supports
 - Fitting contacts
- **Efficient when compared to nonlinear methods**

Theory / Implementation

- The linear gaps provide for simple contact, (compressive) forces only.
- MPC's are used to define the gaps
 - The displacements and mpcforces are used to determine if each gap is “welded” or “free”
 - Displacement criteria: no penetration (MPC displacement ≥ 0)
 - MPCFORCE criteria: compressive force only (MPCforce ≤ 0)
 - An iterative procedure is used to solve the model based on displacements/mpcforces



Opening Grid
Closing Grid
Initial Opening

$U_{y\text{ opening}}$ - positive disp will open the gap
 $U_{y\text{ closing}}$ - positive disp will close the gap
 $U_{y\text{ initial}}$ - initial separation between grids

The equation for the gap displacement can be written as:

$$U_{y\text{ gap}} = U_{y\text{ opening}} - U_{y\text{ closing}} + U_{y\text{ initial}}$$

Rearranging in MPC format and setting $U_{y\text{ closing}}$ as the dependant (first) term:

$$U_{y\text{ closing}} - U_{y\text{ opening}} + U_{y\text{ gap}} - U_{y\text{ initial}} = 0$$

Theory / Implementation

- SPOINTs are defined for U_{gap} and U_{init}
- SUPORT puts SPOINT 55 (U_{gap}) in the R-set for iteration.

SPOINT 55 155

SUPPORT 55 0

- SPC defines initial gap opening = .05 (U_{init})

SPC 101106 155 0 .05

- MPC Entry: (Eq. of motion: $U_{y3} - U_{y2} + U_{gap} - U_{init} = 0$)

\$	SID	G1	C1	A1	G2	C2	A2
MPC	101106	3	2	1.0	2	2	-1.0
\$		G3	C3	A3	G4	C4	A4
		55	0	1.0	155	0	-1.0

MSC.Patran Interface

- **PCL Utility developed to easily create linear gaps and write appropriate MSC.Nastran input entries.**
 - Easy Creation/Deletion of linear gaps
 - Useful graphics for visualization
 - Special Verification
 - Creates complete, run-ready, MSC.Nastran job
 - Stores gaps on MSC.Patran db
 - Session file support

CREATING GAPS

Initial opening defined by nodal distance (automatically calculated) or by user specified value

Gap Opening Distance

Distance: Nodal Distance

Value

- ☒ Magnitude
- ☐ Dx in An. CS
- ☐ Dy in An. CS
- ☐ Dz in An. CS

Node Proximity Tolerance

Gap Opening Distance

Distance: User Value

Initial Gap Opening

.25

Select Nodes (and direction) on both sides of gap, interface will automatically determine "Opening" and "Closing" Nodes and check for SPC/MPC conflicts

Action: Create

Object: Gap

Method: Automatic

Initial Status (DMIG CDSHUT)

☒ Shut ☐ Open

☐ Interference Gap

☐ Gap To Ground

Gap Opening Distance

Distance: Nodal Distance

Value: Magnitude

Node Proximity Tolerance

If Coinc. Node, Opening Node is:

☒ Higher ID ☐ Lower ID

Create Gap if Nodes Closer than:

0.05

☐ Auto Execute

Nodes in Gap

Gap Direction (Analysis C.S.)

☒ Ux ☐ Uy ☐ Uz

Select Nodes in Gaps:

-Apply- Cancel

Linear Gap Elements

Initial Assumption: Opened or Closed

Special Cases

Create gaps within tolerance; if coincident nodes identify the opening node based on Node ID

CREATING GAPS

Initial opening defined by nodal distance (automatically calculated) or by user specified value

Gap Opening Distance
Distance: Nodal Distance
Value: Magnitude
Dx in An. CS
Dy in An. CS
Dz in An. CS

Gap Opening Distance
Distance: User Value
Initial Gap Opening: .25

Action: Create
Object: Gap
Method: Manual

Initial Position (DMIG CDSHUT)
☒ Shut ☐ Open

Gap Opening Distance
Distance: Nodal Distance
Value: Magnitude

☐ Auto Execute

Opening Node
Gap Direction (Analysis C.S.)
☐ Ux ☐ Uy ☒ Uz
Select Opening Nodes:

Closing Node
Gap Direction (Analysis C.S.)
☐ Ux ☐ Uy ☒ Uz
Select Closing Nodes:

-Apply- Cancel

Linear Gap Elements

Initial Assumption:
Opened or Closed

User defines
"Opening" Node(s),
"Closing" Node(s),
and directions.

VISUALIZING GAPS

Group: **Entire Model**
Current Group
Posted Groups

Action: **Display**
Group: **Entire Model**

☒ Auto Update

Gap Direction Vectors
☒ Show Vectors when Selecting
☐ Show Vectors on Gaps
Opening Vector:
Closing Vector:

Independent Node
Shape: **triangle**
1 25
Size of Marker: 10

Dependant Node
Shape: **square**
1 25
Size of Marker: 10

☐ Gap Opening Distance
☐ Initial Position

Update Cancel

Linear Gap Elements

Gap Display: No Options

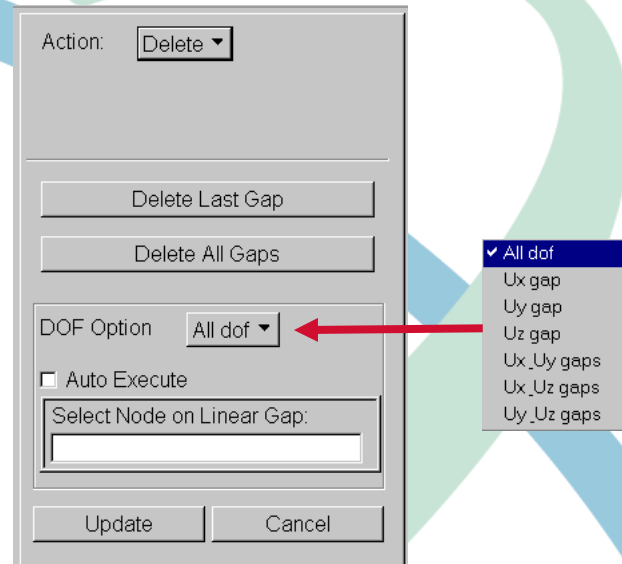
Gap Display: All Options

(0.15-S)

(0.15)

(-S)

DELETING GAPS



CREATING JOB

The image shows two dialog boxes for creating a job in MSC/Nastran. The left dialog is for creating a job, and the right is for creating a bulk data file. A red arrow points from the 'Entire Model' group in the left dialog to the 'Entire Model' group in the right dialog.

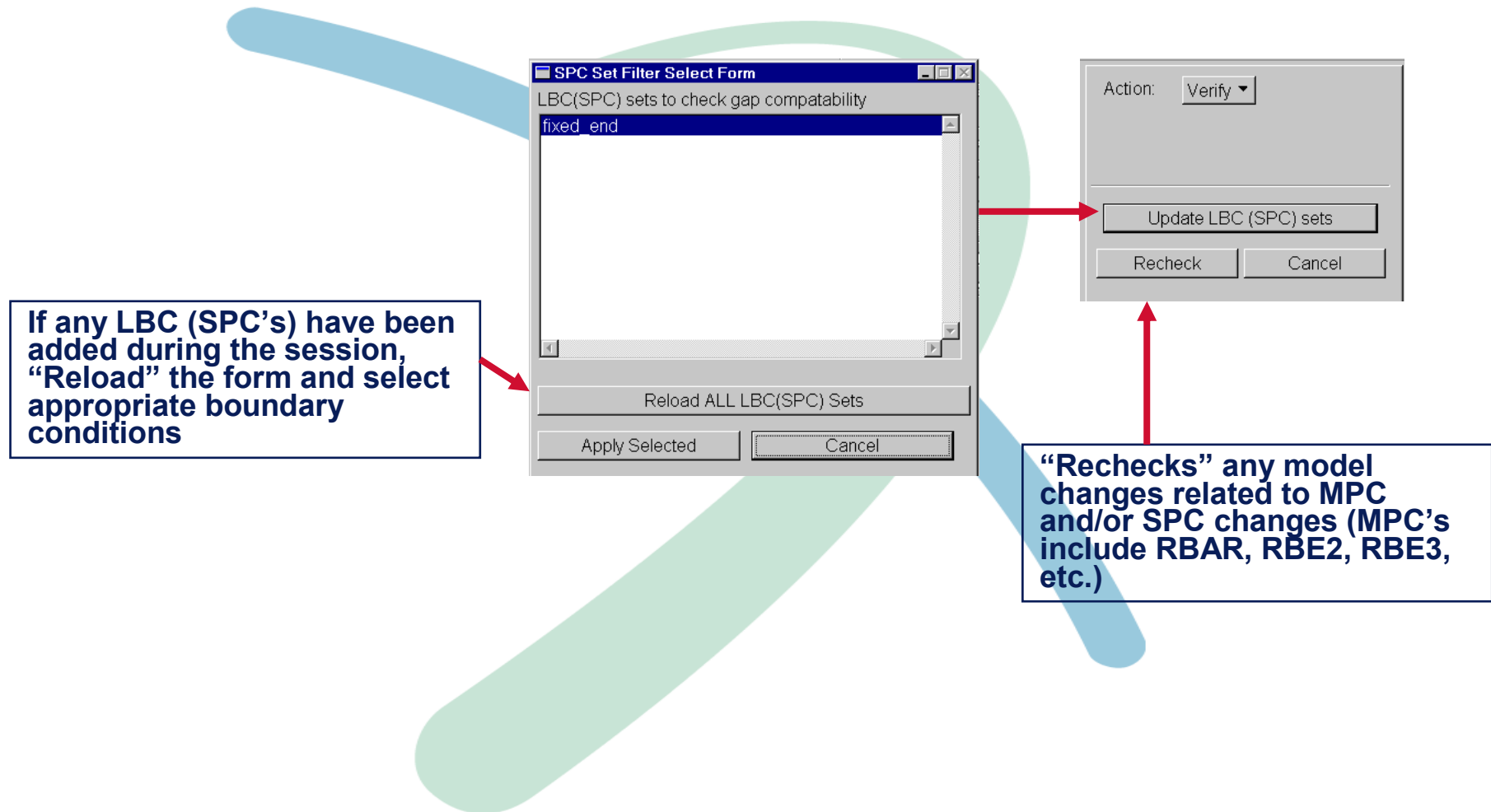
Left Dialog (Job Creation):

- Action: Analysis
- Object: Create
- Method: Job
- Group: Entire Model
- Number of iterations allowed: 1 to 100 (slider)
- PARAM CDITER: 50
- ☒ Param CDPCH
- ☐ Param CDPRT
- Starting Spoint: 1000
- Job Name: lingap
- Buttons: Create, Cancel

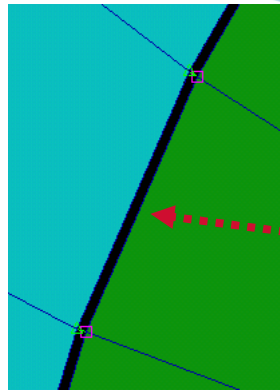
Right Dialog (Bulk Data File Creation):

- Action: Analysis
- Object: Create
- Method: Gap bdf
- Group: Entire Model
- Number of iterations allowed: 1 to 100 (slider)
- PARAM CDITER: 50
- ☒ Param CDPCH
- ☐ Param CDPRT
- Starting Spoint: 1000
- Bulk Data File Select: [Button]
- Current Bulk Data File: [Text Box]
- Buttons: Create, Cancel

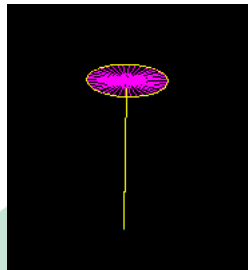
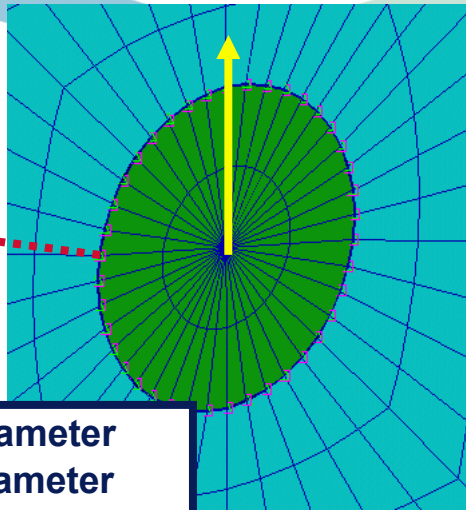
VERIFYING GAPS



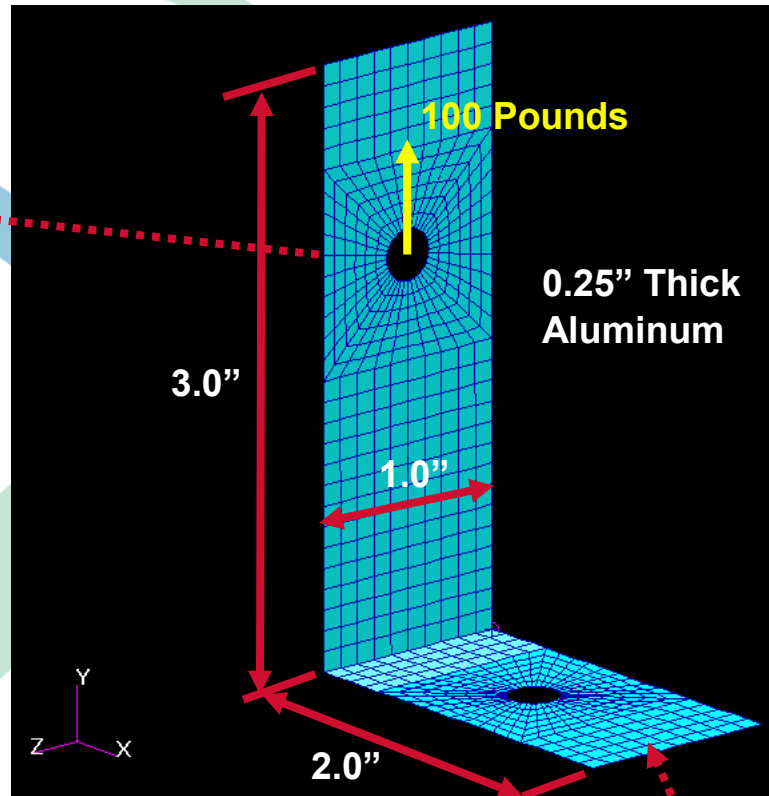
L-Bracket Heel-Toe Analysis



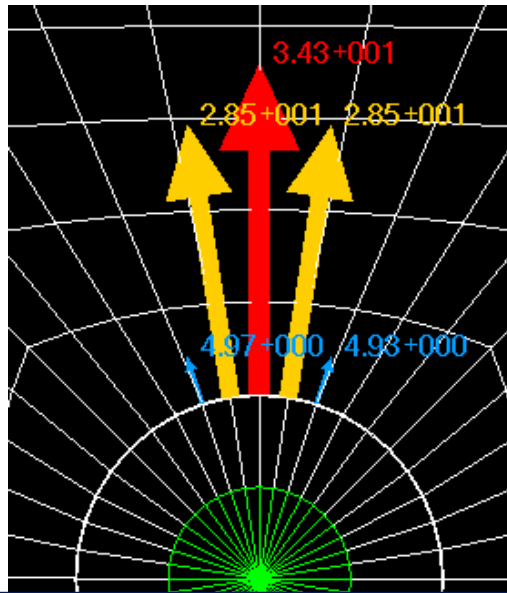
Shear Bolt = .248 Diameter
Shear Hole = .250 Diameter
Linear gaps defined in radial
direction to distribute
bearing load



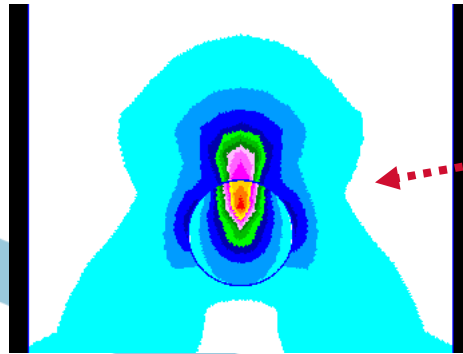
Tension Bolt: 0.5 inches long, .250 Diameter,
attached to plate with RBE3 and fixed at bottom



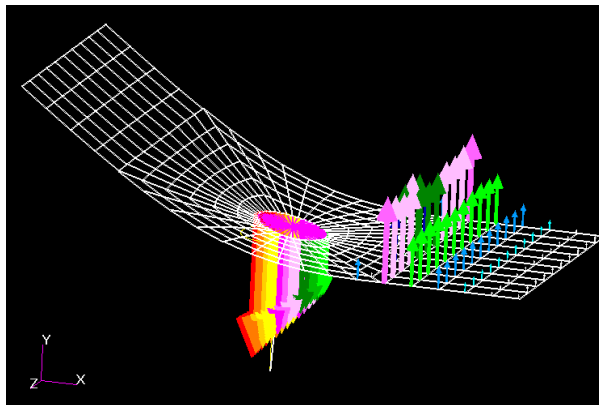
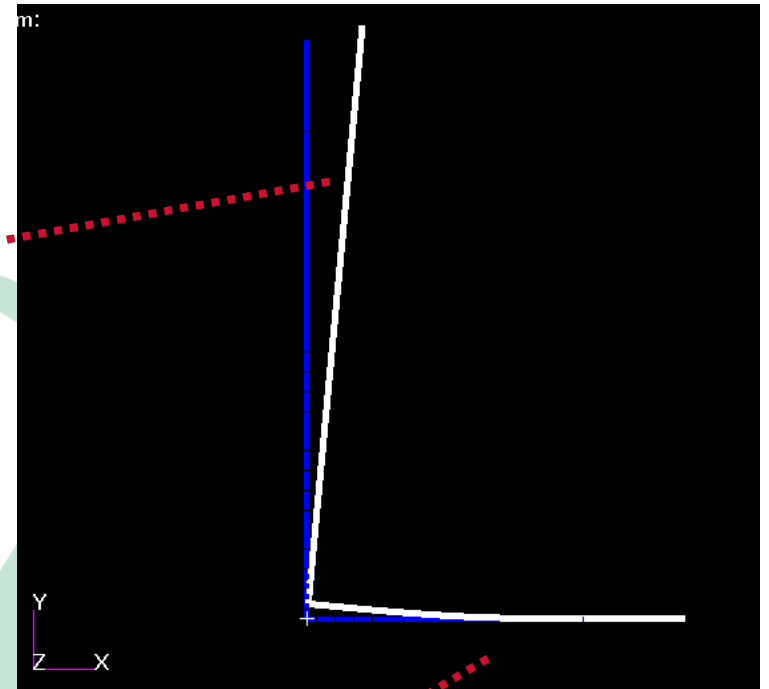
"Gap to Ground" along bottom surface



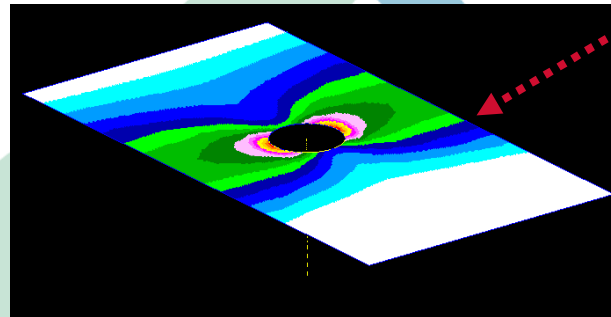
Force Transfer at Shear Bolt



Local Stress Distribution

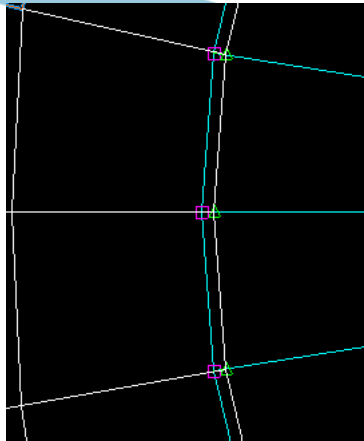


Force Transfer at Tension Bolt

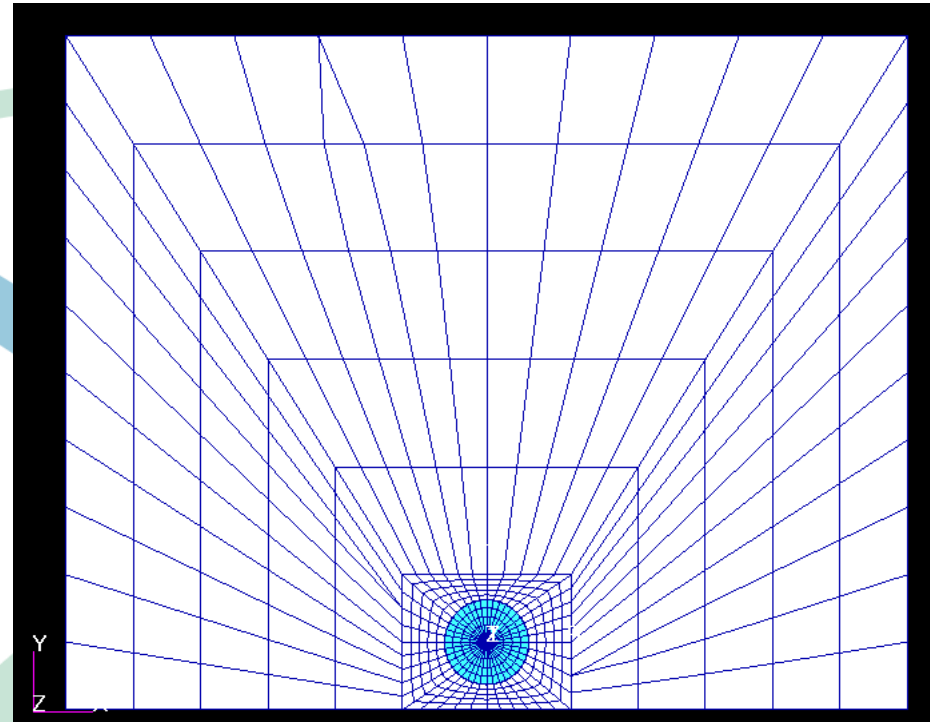


Local Stress Distribution

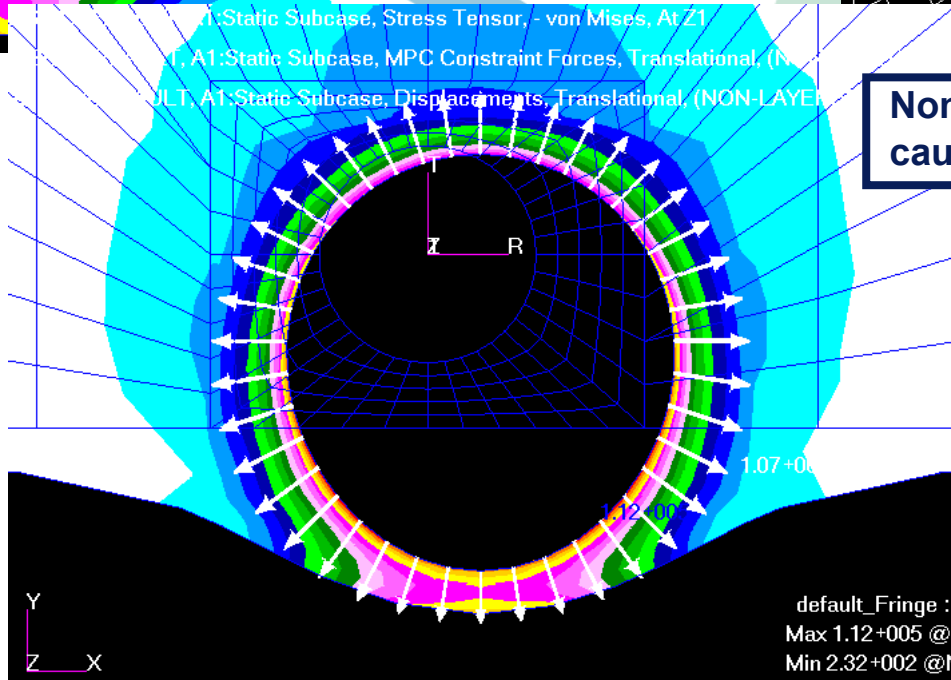
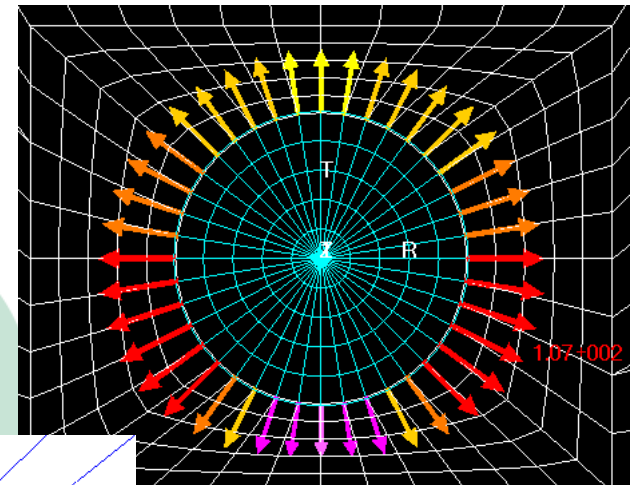
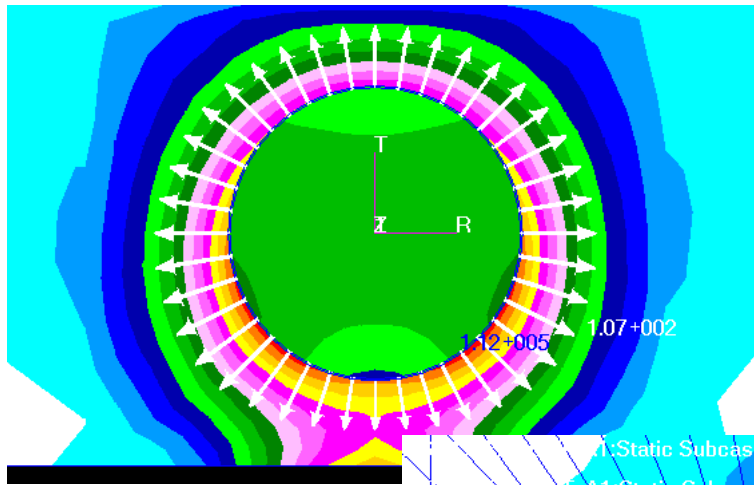
Interference Fit



Interference Gaps:
Opening node is plate
Closing node is pin
Initial Gap determined by
nodal distance



**Plate 5 X 4 X .1
.500 hole .40 from edge
Pin .506 dia X .1 thick
(All Aluminum)**



**Non-uniform force distribution
caused by edge effect**

default_Fringe :
Max 1.12+005 @N
Min 2.32+002 @N

Limitations

- Remember “simple gaps” only
 - NO LARGE DISPLACEMENTS
 - NO FRICTION
 - NO MATERIAL NONLINEARITY
- Otherwise use SOL 106 with CGAP/PGAP
 - with or without PARAM,LGDISP
 - with or without MATS1
 - etc.

Additional Information

- **MSC.Nastran V70.5 Release Guide; Section 3.5, download from:**
<http://www.mechsolutions.com/support/documentation/index.html>
- **McCullough, John and Proctor, Lance, “LOCAL ANALYSIS OF FASTENER HOLES USING THE LINEAR GAP TECHNOLOGY OF MSC/NASTRAN,”**
1999 MSC Aerospace Users’ Conference. Download from:
<http://www.mechsolutions.com/support/library/auc99/>

Conclusion

- Simple gap contact is available in SOL 101 of MSC.NASTRAN V70.5 or later.
 - Constrained Displacement ITERative technique (CDITER).
 - Simple gaps -- no friction
- Works with large models
- Reduces the complexities inherent with SOL 106 (gap stiffness, convergence, etc.)
- Allows use of GPFORCE (now available in SOL 106 in V2001)
- Used in instances where non linearity's do not exist, but contact is required.